Summary

Global sports leagues are uncommon in team sports, as team sports are often only seasonally played at the national level. We are tasked with creating a schedule for the first tournament of this format, choosing Ultimate Frisbee as the sport to develop this Global Sports League (GSL) season schedule for. It is a growing sport that is fairly popular but not widely recognized internationally and that has never been an Olympic sport. We chose 20 teams using criteria that ensured geographic representation and preferred the countries with active Ultimate Frisbee scenes.

Our goal was to create a schedule that is built around a set of factors that are essential to the success of the GSL. We outlined several of these factors, with travel distance and geographic diversity as the most important. Other factors include weather constraints and recovery time. The format of the GSL was then decided heuristically based on the identified factors by taking inspiration from existing tournaments then checking for feasibility. We arrived at a format with two rounds: the preliminary round, where all 20 teams are blocked to minimize travel distance while maintaining diversity, and the playoffs round, where only the top-performing teams advance in order for us to choose a definitive winner by the season's end.

Our solution had three tiers: the blocking, the round robin match-ups, and the actual scheduling. Our strategy was to first balance travel distance and geographic diversity in blocking the teams. Using linear programming – implemented through Python – we were able to arrive at a solution that optimizes the difference between those factors by assigning a "cost" to a match-up. This "cost" weighed the value of geographic diversity by a factor comparable to average distance, and the program preferred a lower cost to a higher cost when grouping teams together. Next, we optimized the matchups within each block to minimize the travel distance of all teams by doing balanced home and away games. The final step of our model was a greedy approach where instead of optimizing for the remaining factors, we fulfilled all the set constraints and spread out the schedule as much as possible. A heuristic approach was used in order to arrive at some values, including the length of the playoffs round. We then had the dates for all matches including the playoffs, so we assigned times per day based on when viewership would increase, keeping in mind the time zones of both home and away teams. The schedule for playoffs was inserted after the preliminary round.

With these conditions set in place, we were able to begin expanding and generalizing our model. The primary factors to consider when doing so were the geographic diversity, revised GSL format, and the new pacing of the matchups. The geographic diversity was determined through the use of weighing the number of associations per continent over the total number globally then taking into account required factors such as the number of active players. Meanwhile, the revised GSL format follows the same blocked round-robin and seeding format with the exception that due to the nature of the values, game determiners together with its required constraints. Finally, the pacing of the matchups were determined by considering the total number of days for both the preliminaries and playoffs together with the corresponding "buffer" times. In doing these, the initial conditions of our model were attained, and the sustainability and feasibility of all factors maximized.

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Letter to the Decision Makers

To the Directors of the IMMC:
Greetings of Good Health!

Your introduction of the Global Sports League (GSL) has opened opportunities for various team sports to publicize themselves in the first international team sports league, and we waste no time in addressing the challenges that team sports will face at the global level. Schedule is undoubtedly the primary form of preparation that may justify your interest, and the potential of your league in itself may be developed by the presentation of a singular sport.

Ultimate Frisbee serves as our client of your consideration.

The Frisbee Advocacy Division (FAD) presents a refined mathematical model that outlines schedules for an Ultimate Frisbee league in a credible seasonal format. The schedule will run from March 1, 2025 to November Firstly, teams will be split into conferences based on geographical distance and continent diversity, which reduces total activity by sharing the number of games in a season among subgroups. Each group will decide their greatest candidates in a regular season round robin format to ensure complete interaction. The top two teams of each group will advance to the playoffs tournament, which is a familiar format in Ultimate Frisbee history. We ensure that this foundation is carefully considered to balance completeness and feasibility.

Secondly, our client is profiled as an outdoor team sport. We ensured fixed matchups through the regular season format, in which every team has played with each of those in the same group. This allows matchup order to be independently organized, which we modelled after time zones and seasonal change. Ultimate games will generate live viewership in the most active times possible for both countries participating, and games will be done in warmer or dryer conditions to prevent health risk and game delay.

Ultimately, everything in our model focuses on a few constraints. We prioritize travel distance and diversity while keeping in mind fairness, recovery and buffer times, weather, and more within the set time frame to ensure that the Global Sports League may run consistently and actively far into the future. We hope this aligns in your interest as a growing institution.

Our mathematical model addresses novel characteristics our client possesses along with the natural discussions that team sports face in a replicable manner. We support the growth and expansion of your institution, and our goal is to allow Ultimate Frisbee to serve as a model for sports of similar caliber. We hope to have developed your spoken challenges into an adequate calculated understanding, and hope for yours as well.

Ultimately Yours,

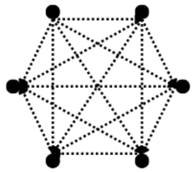
Frisbee Advocacy Division (FAD)

Global Sports League Season Schedule

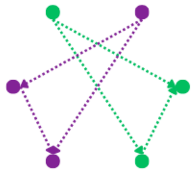
Determining Match-ups

Minimizing Cost: The Trade-Off

Every match-up has an associated cost.

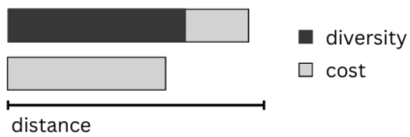


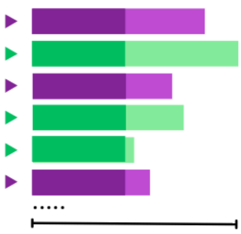
If each team plays against every other team in 190 matches, costs will keep accumulating.



However, if select teams are grouped, costs are significantly reduced.

Cost depends on the travel distance and diversity between the two teams.

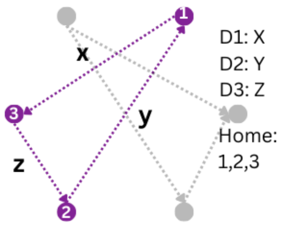




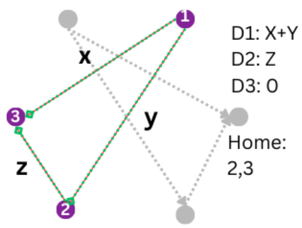
Minimizing costs of blockings can ensure sustainable travels and diverse representation.

Fair Match-ups within Blocks

Given teams with a distance x, y, and z,



VS



To ensure equitable match-ups, home and away teams are designated so that travel distance and home advantage are as equal as possible.

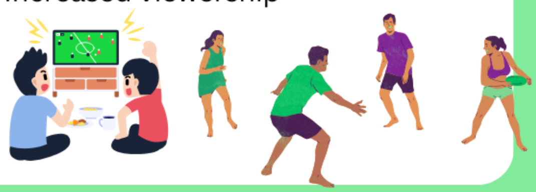
Scheduling Games

A number from 1-168 was assigned to each date allotted for the preliminaries. Certain dates were **off limits** due to constraints.

A systematic program set the pacing for the matches and scheduled them based on the earliest allowed date.

Features

1. Least travel distance
2. Representation of various countries
3. Diverse and convenient matchups
4. Enough recovery time for all teams
5. Accounts for weather and time zones
6. Increased viewership



Dates ← Matchups ← Blockings

Dates → Preliminaries (168 days) → Buffer (14 days) → Playoffs (91 days)

*A match occurs once every **4 days**.*

*A match occurs once every **15 days**.*

40 matches -
starting March 1, 2025

7 matches -
ending Nov 22, 2025

1 | Introduction

1.1 Background

The rise of Ultimate Frisbee is a phenomenon in sports history. Select few modern team sports reach the global level due to complex technical development and lack of rich historical background in colonialism or education. Ultimate, otherwise known as Ultimate Frisbee, which involves throwing a disc between players until it has reached the end zone, defies the normal for both sports origin and popularization. Ultimate has shown that sports created over the past century may also develop audience engagement closer and closer to the global level of sports of rich history like Wrestling and Track & Field, and billionaire-funded team sports like Basketball and Football. Ultimate stands out due to its globalization through mass support and simple, accessible gameplay, without corporate advertisement and history.

Ultimate Frisbee holds several characteristics that highlight its uniqueness as a sport. It is a modern team sport mainly played outdoors. Ultimate is the embodiment of modern sports as it holds ideas such as self-governing characteristics that advocate for the "Spirit of The Game" in a community-built rule book. Team sports require rigid initial technical development as strategic complexity naturally increases with player count, though Ultimate has translated this into high flexibility through minimal equipment and variation brought about by environmental factors. Ultimate upholds the nature of outdoor sports to adjust to environmental factors and incorporate them into gameplay such as the direction and air time of the disc, or the grip required to combat weather. All of these characteristics, however, make Ultimate difficult to schedule as few sports may serve as a reference for it.

Ultimate holds all the challenges that outdoor team sports face at the international level. Firstly, seasonal formats require games to be played year-round, and teams often need much more accommodation and commitment as opposed to individuals. Teams will have prior commitments to existing national associations and international tournaments that are embedded in Ultimate culture. Secondly, schedules become complex due to weather conditions that regulate game quality in the outdoor nature of Ultimate. Players have limited mobility in muddy areas and disc grip loosens in rainy weather. Injuries are at higher risk in the winter season due to stiff muscles and disc texture. Although environmental factors may be considered as natural parts of the game, health and safety are priorities, and conditions that solely hinder player ability must be simply treated as delays to game time.

1.2 Problem Restatement

We are asked to develop a schedule using a mathematical model that will keep in mind certain factors that are important to the Global Sports League (GSL) and the success of the GSL. The GSL is intended to include the very first all-year (spanning 8-9 months) international team sport league, so the schedule cannot be based on existing international sports tournament schedules since there are additional factors to consider, specifically for team sports.

Our first task is to create a schedule that will be most beneficial for our chosen sport and a specific set of 20 teams while accounting for all the identified factors. While doing this, we also need to determine a tournament structure that will be the basis for the schedule so that a league winner can be declared at the end of the 8-9 month season. The next task is to expand this model to fit four additional teams and develop a new schedule for this specific set of 24 teams. This also needs to be tied back to the previously identified factors or constraints. Finally, we move beyond our chosen sport and demonstrate how the model can be modified to fit different team sports since each sport has unique requirements and game lengths that can impact the league schedule.

1.3 Some Term Definitions

Term	Definition
Preliminaries	The first round of the season where all 20 teams participate.
Playoffs	The rounds after the preliminaries where only top performing teams after each round can continue to participate.
Game/match	Used interchangeably to refer to an instance of a team playing against another team.
Blocks/groups	Used interchangeably to refer to the groups of teams playing in the preliminaries.
Round/stage	Used interchangeably to refer to either playoffs or the preliminaries.
Buffer/recovery time	Used interchangeably to refer to gaps between games.

1.4 Assumptions and Justifications

Assumption	Justification
Country associations will be represented by a team.	One metric for choosing the teams is that they have emerging or currently active Ultimate Frisbee scenes, which can be proven by the existence of an Ultimate Frisbee association within the country of origin. Therefore, teams will just be referred to by their country of origin.
Every country that joins the GSL must have a primary international airport.	An international airport is a key requirement for a national association to represent their country in the Global Sports League, as on the global scale, air travel is significantly more efficient.
The locations of teams will be the coordinates of one of their country's major international airports.	This is essential for the sake of simplicity since we are not aware of the exact area of departure nor the airport they will depart from. International airports also often have direct access to one another (from the term "international"), which reduces the consideration of layover flights.
All countries in a region (Northern Hemisphere, Southern Hemisphere, Tropical North, Tropical South) have the same time period in months for seasons (winter and rainy seasons).	This is essential for the sake of simplicity. As the existence or absence of seasons is dependent on the region from the equator, we may assume that these regions will have roughly the same seasonal change as Earth is rotated.

Assumption	Justification
All games will be played in the mixed gender format, in which there is a 3:4 ratio between genders for each team on the field at all times.	The mixed gender format states that there are 4 males and 3 females, or 3 males and 4 females on each team. The initial gender ratio is determined by the additional disc flip, then swapped every two points.
All Ultimate Frisbee matches will be played outdoors.	Ultimate Frisbee requires a large area for players to throw the disc, and on the global scale, the required area is not easily found in the spacial limitations of indoor venues.
Each match will take place within one day.	Ultimate Frisbee matches generally last only a few hours.
All teams will be available to play year-round.	This is essential for the sake of simplicity and not including too many constraints. The availability of 20 teams would be too much to account for in creating the schedule.

1.5 Variables

Variable	Definition
$i,\ j,\ k$	Arbitrary indexes used to label teams.
s	Arbitrary index used to label time slots.
x_{ij}	A binary decision variable. Equal to 1 if team i and team j are in the same group and equal to 0 if otherwise.
y_{ij}	A binary decision variable. Equal to 1 if team i plays against team j and equal to 0 if otherwise.
d_{ij}	Date where team i plays against team j. This will be expressed as an integer that shows the day number from the beginning of the season.
dis_{ij}	The haversine distance in kilometers between team i and team j.
div_{ij}	A binary variable that represents geographical diversity. Equal to 1 if team i and team j are from different countries, and equal to 0 if otherwise.
C_{ij}	The assigned “cost” of grouping team i and team j.
w_{dis}	Weight of travel distance.
w_{div}	Weight of geographic diversity.
lat_i, lon_i	Latitude and longitude of team i.
R	Radius of the Earth.

Variable	Definition
N_c	Total number of participating teams to be sent from a given continent.
n_c	Total number of existing associations for the sport within the continent.
n_t	Total number of existing associations for the sport.
D_t	Desired number of participating teams.
x_a	A binary decision variable. Equal to 1 if a country has one major airport, and equal to 0 otherwise.
P_a	Number of active players in a given association or team.
r	Preliminary ranking of an association before the start of the season.
g_t	The number of groups to block participating teams in prior to the elimination round
T_{games}	Total number of games in a season.
x_t	A variable taking on 3 different variables. Equal to 2 if the 3rd-ranking team wins their first match against the 2nd-ranking team, equal to 1 if the 2nd-ranking team wins the first match against the third ranking team, and equal to 0 if there are no game determiners.
n	Number of teams remaining in the playoffs.
k	Number of preliminary round groups for all participating teams.
b	The duration of a “buffer” period in days.
D_{pr}	Number of days allotted to preliminary rounds.
D_{po}	Number of days allotted to playoff rounds.
D_s	Total number of days allotted for one season.
b_p	The duration of a “buffer” period for a preliminary round.
α	The number of games elapsed since a team last played.
Δ_{ij}	The absolute difference in the respectively chosen UTCs of the home country and the away country.
t_i	The chosen UTC time zone that the given team i’s country of origin belongs to.

2 | Identifying Key Players

2.1 Ultimate Frisbee Teams

The sport we chose is Ultimate Frisbee. In identifying the 20 teams, we decided to take teams from each continent based on the number of existing Ultimate Frisbee associations in that continent, while also ensuring that each continent has at least two teams representing them in the GSL.

The general expression to calculate for this (with the necessary constraints in mind) is given by:

$$N_c = \left[\frac{n_c}{n_t} \left(D_t \right) \right]$$

This equation weighs the number of countries per continent that can be sent – meeting the standards of geographic diversity required. It is the most feasible way to ensure that all continents are equally represented based on their respective means.

Continent	% Total Number of Associations	Number of Teams
Asia	24.79%	4
Africa	17.95%	3
Europe	36.75%	7
North America	5.98%	2
South America	11.97%	2
Oceania	2.56%	2

We used several criteria for choosing the teams, namely geographic representation, urban centers, and the presence of emerging or active Ultimate Frisbee scenes. We did this by choosing teams that cover different countries of each continent and that come from cities with a large population that have access to international airports. The top associations within a continent were determined by active player base count[9].

In general, we calculated this as:

$$r = x_a + P_a$$

The teams and their locations are as follows:

Europe	Asia
1. Deutscher Frisbeesport-Verband (DFV) - Germany Frankfurt Airport - Latitude: 50.0379, Longitude: 8.5622. 2. UK Ultimate (UKU) - United Kingdom London Heathrow Airport - Latitude: 51.470020, Longitude: -0.454295 3. Fédération Française de Flying Disc (FFFD) - France	8. Japan Flying Disc Association (JFDA) - Japan Tokyo Haneda Airport - Latitude: 35.553333, Longitude: 139.781113 9. Chinese Flying Disc Administrative Center (CFDAC) – China Beijing Daxing International Airport - Latitude: 39.50917, Longitude: 116.41056 10. Korea Ultimate Players Association (KUPA) – South Korea Seoul Incheon International Airport - Latitude: 37.463333, Longitude: 126.440002

Paris Charles de Gaulle Airport - Latitude: 49.009724, Longitude: 2.547778 4. FIGeST Federazione Italiana Giochi e Sport Tradizionali (FIG) - Italy Rome Fiumicino Airport - Latitude: 41.773540, Longitude: 12.239712 5. Nederlandse Frisbee Bond (NFB) - Netherlands Amsterdam Airport Schiphol - Latitude: 52.308056, Longitude: 4.764167 6. Swedish Frisbeesport Federation (SFF) - Sweden Stockholm Arlanda Airport - Latitude: 59.65015, Longitude: 17.94362 7. Österreichischer Frisbee-Sport Verband (ÖFSV) - Austria	11. Ultimate Players Association of India (UPAI) – India Indira Gandhi International Airport - Latitude: 28.556160, Longitude: 77.100281
Africa	**North America**
12. South African Flying Disc Association (SAFDA) - South Africa Tambo International Airport - Latitude: -26.134789, Longitude: 28.240528 13. Kenya Flying Disc Association (KFDA) - Kenya Jomo Kenyatta International Airport - Latitude: -1.333731, Longitude: 36.927109 14. Uganda Flying Disc Federation (UFDF) - Uganda Entebbe International Airport - Latitude: 0.044721, Longitude: 32.443055	15. USA Ultimate (USAU) Denver International Airport - Latitude: 39.849312, Longitude: -104.673828 16. Ultimate Canada (UC) Toronto Pearson International Airport - Latitude: 43.677128, Longitude: -79.633453
South America	**Oceania**
17. Federación Colombiana de Disco Volador (FECODV) - Colombia El Dorado International Airport - Latitude: 4.7008, Longitude: -74.1415. 18. Federacio Paulista de Disco (FPD) - Brazil São Paulo–Guarulhos International Airport - Latitude: -23.4258, Longitude: -46.4683.	19. Australian Flying Disc Association Inc. (AFDA) - Australia Sydney Airport - Latitude: -33.947346, Longitude: 151.179428. 20. New Zealand Ultimate (NZU) - New Zealand Auckland Airport - Latitude: -37.0048, Longitude: 174.7883.

2.2 Factors for the Schedule

Through research, we decided on ten factors that would affect the season league schedule. These will act as constraints or goals that the schedule must fulfill in order to ensure the success of the GSL.

1. Travel distance
   This is directly related to the time and money that teams would spend traveling to different parts of the world for matches. No team should have to travel significantly more than another team because that would be unfair since travel can cause fatigue and additional expenses. Additionally, this metric should be minimized as much as possible to maintain the logistical feasibility of the GSL and lessen the carbon footprint from plane rides. However, this could compromise the diversity of matches, since some teams may not be able to travel to areas that are too far. This factor is an umbrella for economic and environmental sustainability since it is the thing that affects these aspects the most.
2. Geographic diversity
   Teams should not be primarily matched up with teams in the same continent as them to increase the diversity of matches. The trade-off for this is the travel distance, because matching up teams from completely different continents means they will have to travel further for games.
3. Number of games played

As much as possible, the number of games played by each team should be equal. At the very least, the format of the league should give each team equal opportunity to be the overall winner.

4. Equitable matchups
   The matchups should be spread out evenly as much as possible. This means that within a group, each team should play each of the other teams at least once. When assigning matchups in the case of playoffs, seeds should be considered. This is to ensure fairness and keep the GSL competitive.
5. Home and away game balance
   Each team should play an equal number of home and away games. Having it otherwise can cause some teams to have a home advantage, and be unfairly taxing on the visiting team due to travel. If this is strictly imposed, the total travel expenses may increase because each team will need to travel for away games.
6. Time zone differences
   Since this is an international team sports league, the dates and times of matches need to be dependent on the time zone differences of the team locations. This would prevent jet lag but create more scheduling constraints because the games need to be held at appropriate dates and times for both participating teams.
7. Recovery time
   The teams will need ample recovery time in between matches for the sake of athlete health and performance. This time should not include travel time.
8. Weather constraints
   Ultimate Frisbee is a primarily outdoor sport that requires a field of grass or artificial grass. Not every venue will have an indoor field with artificial grass, so to ensure the flexibility of venues, the weather needs to be considered. Games cannot be held during the winter nor during rainy seasons (to lessen the chance of cancellation of games due to rain) of certain areas.
9. Adaptability
   The schedule should be able to accommodate changes in input. It should also still perform well under stress or disruption. For example, if a team drops out or there are travel difficulties, the changes in the schedule should be minimal. Ensuring adaptability, however, can result in the need for more buffer time and fewer games per season.
10. Viewership
    The schedule, along with the matchups, should be able to maximize the viewership yield to ensure the sustainability of the GSL along with the net profit. Thus, it is vital that the pacing of an entire season's matchups maximizes audience viewership while keeping their interest intact and increasing.

3 | Developing the Initial Schedule

3.1 GSL Season Format

3.1.1 Overview

The season will begin with a set of matches between teams that are blocked into four groups of five teams each. After each team has played every other team in their group once (round robin-style), the top two teams from each group based on number of matches won will rise to the playoffs stage for a total number of eight participating teams. The winners of each match will play against each other until only one team, the league winner, remains. In total, there will be 47 games played over the course of nine months.

In the preliminary round where the teams are blocked, each team will play four games, one against each team in the same group: two home games and two away games. This fulfills the home-away game balance and also allows for equitable matchups.

For the playoffs stage, the locations and matchups are still to be determined based on performance during the preliminary stage. The top two teams with the most wins will advance to this round. However, they will be seeded (a common practice in tournaments for viewership because it increases the chances of the best teams facing each other only in the final round) or ranked among each other based on their previous perceived performance, and this will determine the draws or matchups. In the quarterfinals, the Rank 1 team will go against the Rank 8 team, the Rank 2 team will go against the Rank 7 team, and so on. It follows this general format for any team i ranked ith (the match index also determines the order of play):

$$\text{Match } i\text{: team } i \text{ vs team } (n + 1 - i)$$

The location of each playoff match also depends on the matchups. The home team will be the lower ranked team while the away team will be the higher ranked team, for fairness.

3.1.2 Format Justification

This game format was chosen due to three main factors – its feasibility, sustainability, and game probability. In the problem-solving process, we identified two possible game formats. The first being a format where every team plays every other participating team once and the second being our chosen format. Both directions are seemingly feasible, but ultimately, the benefits of our chosen format outweighed that of the other.

First, in terms of sustainability, this format favors this factor more as it reduces the amount of games played by a large margin while still ensuring that matchups are equitable and properly distributed. In this regard, had we chosen the first format, the number of games to be played within a season would be 190. On the other hand, with our format, this number goes down by over 75% through the use of blocking. Because of this, teams also have to travel less with the reduced number of games. This lessens both the cost burden on teams and carbon impact to the earth. The games are still equitable, however, as teams are chosen on strictly followed factors, and the execution of the games follow a strict policy to ensure the fairness of the games.

In line with the sustainability, the feasibility of the games along with the entire schedule becomes much more probable given the eight to nine month season. This is because more “buffer” times are provided to ensure that all factors and conditions are properly lined up and organized. Rather than having to follow a more cramped schedule with more games to execute, the presence of less games gives time to properly fix any errors or inconveniences that occur during the season. Additionally, with feasibility and unexpected inconveniences, it also provides time to rectify any games that end in ties. In this regard, ties can be addressed and properly organized during the season.

Finally, along with game probability, less games correlate to less ties. Since less matches are to be played, this means that there is a smaller chance that ties will occur — a factor highly beneficial when considering the feasibility of our study. An important thing to note though is that this format is still highly capable of reaching the same outcome with reduced games. This is because both the “weak” and “strong” teams are “captured” efficiently meaning that the weaker teams will still be eliminated early on with this format while the stronger teams are able to qualify to the following rounds where the format becomes seeded. While there may be some limitations with our model — especially when considering the possibility of two or more of the stronger teams being assigned in one group — this feasibility of this format lies in the fact that at least two or more teams per group can be chosen. In doing this, the full seeded playoffs can commence, and ultimately, the winner declared. Thus, our format, considering all three factors, is an optimal choice leading to a better flow of the GSL season.

3.2 Initial Schedule

3.2.1 Blocking the Teams

Travel distance and geographic diversity are the two factors that are trade-offs for the other. This means we need to find a balance between them in the blockings through optimizing. We want to form four groups of five teams such that the travel distance is minimized and the geographic diversity is maximized. We will do this through assigning a “cost” to grouping two teams, team i and team j. A high cost for a pair of teams means the two teams will preferably not be grouped together while a low cost means they will likely be grouped together. The cost is defined as:

$$C_{ij} = w_{dis} \cdot dis_{ij} - w_{div} \cdot div_{ij}$$

where

$$dis_{ij} = 2R \cdot arctan2(\sqrt{a}, \sqrt{1 - a})$$

for

$$R = 6371 \text{ (the radius of the Earth)}$$

$$a = sin^2\left(\frac{radians(lat_j - lat_i)}{2}\right) + cos(radians(lat_i)) \cdot cos(radians(lat_j)) \cdot sin^2\left(\frac{radians(lon_j - lon_i)}{2}\right)$$

This is based on the haversine formula for computing the distance between two points on a sphere (in this case, the Earth) using the latitude and longitude of those two points.

Additionally,

$$w_s = 1$$

$$w_v = 10229.43$$

The weight on diversity is to express the "value" of diversity to something comparable to travel distance. The number was calculated by taking the average intercontinental distance (calculated using the code in Appendix [1]) between all the teams in order to balance the prioritization of travel distance and geographic diversity.

We want to minimize the total cost over all chosen $x_{ij} = 1$.

Our goal is to minimize

$$\sum_{i<j} (C_{ij} \cdot x_{ij})$$

or

$$\sum_{i<j} ([w_{dis} \cdot dis_{ij} - w_{div} \cdot div_{ij}] \cdot x_{ij})$$

under the constraint that every group must have five teams each using the following set of equations:

$$\sum_{j \neq i} x_{ij} = 4 \text{ for all } i$$

$$x_{ij} = x_{ji}$$

$$x_{ij} + x_{jk} - 1 \leq x_{ik}$$

$$x_{ik} + x_{kj} - 1 \leq x_{ij}$$

$$x_{ji} + x_{ik} - 1 \leq x_{jk}$$

Using the code in Appendix [1], the groups that will give the most optimal travel distance and diversity are listed below with the team name, country of origin, continent, and coordinates:

Group 1	Group 2
• DFV (Germany) (Europe, lat=50.0379, lon=8.5622) • SFF (Sweden) (Europe, lat=59.6501, lon=17.9436) • OFSV (Austria) (Europe, lat=48.1108, lon=16.5708) • UPAI (India) (Asia, lat=28.5562, lon=77.1003) • KFDA (Kenya) (Africa, lat=-1.3337, lon=36.9271)	• UKU (UK) (Europe, lat=51.4700, lon=-0.4543) • NFB (Netherlands) (Europe, lat=52.3081, lon=4.7642) • USAU (USA) (North America, lat=39.8493, lon=-104.6738) • UC (Canada) (North America, lat=43.6771, lon=-79.6335) • FECODV (Colombia) (South America, lat=4.7008, lon=-74.1415)
Group 3	**Group 4**
• FFFD (France) (Europe, lat=49.0097, lon=2.5478) • FIG (Italy) (Europe, lat=41.7735, lon=12.2397) • SAFDA (South Africa) (Africa, lat=-26.1348, lon=28.2405) • UFDF (Uganda) (Africa, lat=0.0447, lon=32.4431) • FPD (Brazil) (South America, lat=-23.4258, lon=-46.4683)	• JFDA (Japan) (Asia, lat=35.5533, lon=139.7811) • CFDAC (China) (Asia, lat=39.5092, lon=116.4106) • KUPA (South Korea) (Asia, lat=37.4633, lon=126.4400) • AFDA (Australia) (Oceania, lat=-33.9473, lon=151.1794) • NZU (New Zealand) (Oceania, lat=-37.0048, lon=174.7883)

3.2.2 Group Matchups

Within each group, we need to decide matchups while minimizing the distance traveled.

Our goal is to minimize

$$D = \sum_{i<j} (dis_{ij} \cdot y_{ij})$$

while ensuring that each team plays exactly two home games and two away games,

$$\sum_{j \neq i} y_{ij} = 2 \text{ for each team } i$$

$$\sum_{j \neq i} y_{ji} = 2 \text{ for each team } i$$

where y_{ij} represents a match where team i plays away against team j and y_{ji} represents a match where team i plays home against team j. We also need to ensure that each team in a group plays against every other team in the same group exactly once for a total of four games. We do this by dictating that they must either play at team i's home or team j's home, but not both:

$$y_{ij} + y_{ji} = 1 \text{ for all pairs of teams } i, j$$

Using these constraints in the code in Appendix [2], we arrive at the following matchups:

Group 1	Group 2
• SFF (Sweden) (Home) vs DFV (Germany) (Away) • DFV (Germany) (Home) vs OFSV (Austria) (Away) • UPAI (India) (Home) vs DFV (Germany) (Away) • DFV (Germany) (Home) vs KFDA (Kenya) (Away) • OFSV (Austria) (Home) vs SFF (Sweden) (Away) • SFF (Sweden) (Home) vs UPAI (India) (Away) • KFDA (Kenya) (Home) vs SFF (Sweden) (Away) • UPAI (India) (Home) vs OFSV (Austria) (Away) • OFSV (Austria) (Home) vs KFDA (Kenya) (Away) • KFDA (Kenya) (Home) vs UPAI (India) (Away)	• NFB (Netherlands) (Home) vs UKU (UK) (Away) • USAU (USA) (Home) vs UKU (UK) (Away) • UKU (UK) (Home) vs UC (Canada) (Away) • UKU (UK) (Home) vs FECODV (Colombia) (Away) • USAU (USA) (Home) vs NFB (Netherlands) (Away) • UC (Canada) (Home) vs NFB (Netherlands) (Away) • NFB (Netherlands) (Home) vs FECODV (Colombia) (Away) • UC (Canada) (Home) vs USAU (USA) (Away) • FECODV (Colombia) (Home) vs USAU (USA) (Away) • FECODV (Colombia) (Home) vs UC (Canada) (Away)
Group 3	**Group 4**
• FIG (Italy) (Home) vs FFFD (France) (Away) • SAFDA (South Africa) (Home) vs FFFD (France) (Away) • FFFD (France) (Home) vs UFDF (Uganda) (Away) • FFFD (France) (Home) vs FPD (Brazil) (Away) • SAFDA (South Africa) (Home) vs FIG (Italy) (Away) • UFDF (Uganda) (Home) vs FIG (Italy) (Away) • FIG (Italy) (Home) vs FPD (Brazil) (Away) • UFDF (Uganda) (Home) vs SAFDA (South Africa) (Away) • FPD (Brazil) (Home) vs SAFDA (South Africa) (Away) • FPD (Brazil) (Home) vs UFDF (Uganda) (Away)	• CFDAC (China) (Home) vs JFDA (Japan) (Away) • KUPA (South Korea) (Home) vs JFDA (Japan) (Away) • JFDA (Japan) (Home) vs AFDA (Australia) (Away) • JFDA (Japan) (Home) vs NZU (New Zealand) (Away) • KUPA (South Korea) (Home) vs CFDAC (China) (Away) • AFDA (Australia) (Home) vs CFDAC (China) (Away) • CFDAC (China) (Home) vs NZU (New Zealand) (Away) • AFDA (Australia) (Home) vs KUPA (South Korea) (Away) • NZU (New Zealand) (Home) vs KUPA (South Korea) (Away) • NZU (New Zealand) (Home) vs AFDA (Australia) (Away)

3.2.3 Accounting for Other Factors

Now that we have the matchups, the exact dates and time of each match need to be determined. The following constraints will need to be fulfilled:

- Recovery time: No team should have two games within two weeks of each other (each team should play at most every other week).
- Weather constraints for home games, determined based on seasons:
    - Teams in the Northern Hemisphere can't play from December to March (winter).

- Teams in the Southern Hemisphere can't play from July to December (winter).
- For tropical countries:
  - Teams in the Northern Hemisphere can't play from May to July (rainy season).
  - Teams in the Southern Hemisphere can't play from November to February (rainy season).

- <u>Viewership:</u> No two matches should happen simultaneously.
- <u>Time zone differences:</u> Match times should be between the local time of 8 am to 8 pm for both teams.
- <u>Adaptability:</u> There should be ample buffer times between matches.

The season lasts from eight to nine months, for a maximum of 39 weeks or 273 days. We propose that the schedule be stretched out as much as possible throughout the 273 days in order to factor in adaptability using buffer periods and also increase viewership If travel delays or unseen complications occur, there will still be several days in between each match to account for these. There are two stages in the GSL season format, the preliminary blocked rounds and the playoffs, for a total of 47 games (40 from the preliminaries and seven from the playoffs). In between the preliminary rounds and the playoffs, there should be a buffer period of 14 days for the advancing teams to prepare.

We will employ a heuristic method in scheduling the playoffs matches. There will be 14 days between matches and six such periods coming to a total of $14 + 7 + (14 \cdot 6) = 105$ days, leaving $273 - 105 = 168$ days for the preliminary stage. For the preliminary stage, we assign every date a number from 1 to 168 starting from March 1, 2025 (1) to August 15, 2025 (168). These months were chosen to give a large allocation for regions with a lot of matches, and also avoid holidays. The following table shows the available dates (in green) for each region under the weather constraints:

Region (# of matches held)	March	April	May	June	July	August	Available dates
Northern Hemisphere (24)							32-168
Southern Hemisphere (6)							1-122
Tropical North (8)							1-61, 154-168
Tropical South (2)							1-168

There are 168 days for 40 matches, so each match will occur every $168 \div 40 \approx 4$ days (three days in between each match). The minimum recovery time for each team is 14 days, so each team can only play every four matches. In addition to this, there should only be one match per day.

Within a day, viewers are most likely to watch between 8 am and 8 pm. Assuming games (the match and the events) last around two hours, the game must start between 8am and 6pm in the local times of both countries. To account for time zone differences,

$$\Delta_{ij} = |t_j - t_i|$$

there is a range of possible times in military time for the match to start given by

$$[8 + \Delta_{ij}, 18] \text{ (for the country with } max(t_i, t_j))$$

$$[8, 18 - \Delta_{ij}] \text{ (for the country with } min(t_i, t_j))$$

where the average of these ranges are chosen:

$$\frac{26+\Delta_{ij}}{2} \text{ (for the country with } max(t_i, t_j))$$
$$\frac{26-\Delta_{ij}}{2} \text{ (for the country with } min(t_i, t_j))$$

Using the code in Appendix [6], we were able to arrive at a schedule (refer to the Appendix to view the output). For each matchup going from top to bottom, the program attempts to find the first unscheduled match that fits the weather constraint window for the match's region and satisfies the required recovery time for both teams. It employs a "greedy" mechanism that simply takes the first satisfactory slot rather than optimizing the schedule to have the least variance in recovery time for all teams (we considered this, but there was no definitive solution found since there are many schedules that satisfy the conditions). The order in which the program checks the matchups was shuffled to ensure all matches were allotted. The final initial schedule (including the playoffs) is shown below:

Match	Date	Day	Location	Away Team Country	Kickoff Time (Home)	Away Time
UFDF (Uganda) vs SAFDA (South Africa)	Mar 1, 2025	Sat	Uganda	South Africa	1:30 PM	12:30 PM
AFDA (Australia) vs KUPA (South Korea)	Mar 5, 2025	Wed	Australia	South Korea	1:30 PM	12:30 PM
UPAI (India) vs DFV (Germany)	Mar 9, 2025	Sun	India	Germany	2:45 PM	11:15 AM
KFDA (Kenya) vs SFF (Sweden)	Mar 13, 2025	Thu	Kenya	Sweden	1:30 PM	12:30 PM
FECODV (Colombia) vs USAU (USA)	Mar 17, 2025	Mon	Colombia	USA	1:30 PM	12:30 PM
NZU (New Zealand) vs KUPA (South Korea)	Mar 21, 2025	Fri	New Zealand	South Korea	2:30 PM	11:30 AM
AFDA (Australia) vs CFDAC (China)	Mar 25, 2025	Tue	Australia	China	2:00 PM	12:00 PM
UPAI (India) vs OFSV (Austria)	Mar 29, 2025	Sat	India	Austria	2:45 PM	11:15 AM
SFF (Sweden) vs DFV (Germany)	Apr 2, 2025	Wed	Sweden	Germany	1:00 PM	1:00 PM
NFB (Netherlands) vs UKU (UK)	Apr 6, 2025	Sun	Netherlands	UK	1:30 PM	12:30 PM
NZU (New Zealand) vs AFDA (Australia)	Apr 10, 2025	Thu	New Zealand	Australia	2:00 PM	12:00 PM
OFSV (Austria) vs KFDA (Kenya)	Apr 14, 2025	Mon	Austria	Kenya	12:30 PM	1:30 PM
SFF (Sweden) vs UPAI (India)	Apr 18, 2025	Fri	Sweden	India	11:15 AM	2:45 PM
USAU (USA) vs UKU (UK)	Apr 22, 2025	Tue	USA	UK	9:30 AM	4:30 PM
UC (Canada) vs NFB (Netherlands)	Apr 26, 2025	Sat	Canada	Netherlands	10:00 AM	4:00 PM
DFV (Germany) vs OFSV (Austria)	Apr 30, 2025	Wed	Germany	Austria	1:00 PM	1:00 PM
FIG (Italy) vs FFFD (France)	May 4, 2025	Sun	Italy	France	1:00 PM	1:00 PM

UKU (UK) vs FECODV (Colombia)	May 8, 2025	Thu	UK	Colombia	4:00 PM	10:00 AM
USAU (USA) vs NFB (Netherlands)	May 12, 2025	Mon	USA	Netherlands	9:00 AM	5:00 PM
DFV (Germany) vs KFDA (Kenya)	May 16, 2025	Fri	Germany	Kenya	12:30 PM	1:30 PM
OFSV (Austria) vs SFF (Sweden)	May 20, 2025	Tue	Austria	Sweden	1:00 PM	1:00 PM
UKU (UK) vs UC (Canada)	May 24, 2025	Sat	UK	Canada	3:30 PM	10:30 AM
NFB (Netherlands) vs FECODV (Colombia)	May 28, 2025	Wed	Netherlands	Colombia	4:30 PM	9:30 AM
SAFDA (South Africa) vs FFFD (France)	Jun 1, 2025	Sun	South Africa	France	1:00 PM	1:00 PM
FIG (Italy) vs FPD (Brazil)	Jun 5, 2025	Thu	Italy	Brazil	3:30 PM	10:30 AM
UC (Canada) vs USAU (USA)	Jun 9, 2025	Mon	Canada	USA	2:00 PM	12:00 PM
CFDAC (China) vs JFDA (Japan)	Jun 13, 2025	Fri	China	Japan	12:30 PM	1:30 PM
FFFD (France) vs UFDF (Uganda)	Jun 17, 2025	Tue	France	Uganda	12:30 PM	1:30 PM
SAFDA (South Africa) vs FIG (Italy)	Jun 21, 2025	Sat	South Africa	Italy	1:00 PM	1:00 PM
KUPA (South Korea) vs JFDA (Japan)	Jun 26, 2025	Thu	South Korea	Japan	1:00 PM	1:00 PM
FFFD (France) vs FPD (Brazil)	Jun 30, 2025	Mon	France	Brazil	3:30 PM	10:30 AM
CFDAC (China) vs NZU (New Zealand)	Jul 4, 2025	Fri	China	New Zealand	11:00 AM	3:00 PM
JFDA (Japan) vs AFDA (Australia)	Jul 9, 2025	Wed	Japan	Australia	12:30 PM	1:30 PM
FPD (Brazil) vs SAFDA (South Africa)	Jul 13, 2025	Sun	Brazil	South Africa	10:30 AM	3:30 PM
KUPA (South Korea) vs CFDAC (China)	Jul 17, 2025	Thu	South Korea	China	1:30 PM	12:30 PM
JFDA (Japan) vs NZU (New Zealand)	Jul 22, 2025	Tue	Japan	New Zealand	11:30 AM	2:30 PM
FPD (Brazil) vs UFDF (Uganda)	Jul 26, 2025	Sat	Brazil	Uganda	10:00 AM	4:00 PM
KFDA (Kenya) vs UPAI (India)	Aug 1, 2025	Fri	Kenya	India	11:45 AM	2:15 PM
FECODV (Colombia) vs UC (Canada)	Aug 5, 2025	Tue	Colombia	Canada	12:30 PM	1:30 PM
UFDF (Uganda) vs FIG (Italy)	Aug 9, 2025	Sat	Uganda	Italy	1:30 PM	12:30 PM
Quarterfinal 1	Aug 24, 2025	Sun	TBD	TBD	TBD	TBD
Quarterfinal 1	Sep 8, 2025	Mon	TBD	TBD	TBD	TBD
Quarterfinal 1	Sep 23, 2025	Tue	TBD	TBD	TBD	TBD
Quarterfinal 1	Oct 8, 2025	Wed	TBD	TBD	TBD	TBD
Semifinal 1	Oct 23, 2025	Thu	TBD	TBD	TBD	TBD
Semifinal 2	Nov 7, 2025	Fri	TBD	TBD	TBD	TBD
Final (Championship)	Nov 22, 2025	Sat	TBD	TBD	TBD	TBD

4 | Expanding the Model

4.1 New Ultimate Frisbee Teams

Four more Ultimate Frisbee teams further diversifies and increases the scope of GSL as a whole. Thus, to keep the standards and feasibility of the competition, these teams must undergo the same process as the initial twenty teams. To begin with, the quantity of the teams originating from the same continent will be determined by reweighing the number of required teams per continent.

Continent	% Total Number of Associations	Number of Teams
Asia	24.79%	**5**
Africa	17.95%	**4**
Europe	36.75%	**8**
North America	5.98%	2
South America	11.97%	**3**
Oceania	2.56%	2

The table above bolds the values of the continents with an increased number of teams representing their area. For this simulation, it was found that the number of teams per continent simply increased by one within these areas. Using this along with the criteria previously utilized in 3.1, the four additional teams were determined to be as follows:

<u>Asia:</u>
Philippine Flying Disc Association (PFDA) - Philippines
Ninoy Aquino International Airport - Latitude: 14.556586, Longitude: 121.023415.
<u>Africa:</u>
Nigeria Flying Disc Sport Association (NFDSA) - Africa
Murtala Muhammed International Airport - Latitude: 6.577369, Longitude: 3.321156
<u>Europe:</u>
Danish Flying Disc Federation (DFDF) - Denmark
Copenhagen Airport - Latitude: 55.620750, Longitude: 12.650462.
<u>South America:</u>
Asociación de Deportes de Disco Volador de la República Argentina (ADDVRA) - Argentina
Ministro Pistarini International Airport - Latitude: -34.822222, Longitude: -58.535833

4.2 Changes to the Schedule

4.2.1 Geographic Representations and Equitable Matchups

The geographic representations of all teams remain equitably represented throughout the season because each continent's contingent is properly chosen and considered thus allowing the diversity of each part to take form. Each continent is properly designated with its prescribed country teams based on the reasonable criteria leading to a complete geographic representation – one in line with the GSL's goals. Additionally, since all teams are properly weighted per continent, the matchups do not become a problem. The matchups are still able to follow the same model discussed in (3) which properly ensures that locations, time zones, and buffer periods are properly represented and taken into account. Essentially, in terms of geographic representation, this has no major impact on equity. Because of this, in terms of geography, this schedule is reasonable. All teams still have to account for equal "costs" leading to the season's execution.

4.2.2 Sustainability

Similar to how the model discussed in (3) better promotes sustainability and practicality across all its game matches, the context of this new format follows suit. Despite having more teams along with more games, this GSL still promotes sustainability as it finds the most optimal and feasible way to find the best teams to compete with each other based on their travel time, location (through coordinates), and their overall performance leading to a successful season.

4.3 Changes to the GSL Season Format

With the addition of four new teams, the number of games to be played by each team and in total will increase by some amount. Nonetheless, the flow of the season format shall stay relatively the same with a few modifications.

Since the total number of teams is still divisible by four, the total number of groups will still remain at four. Thus, each group will have six teams total with each of the six teams playing once against the other five teams. This implies that the total number of games played will be 15, and for the entire preliminary round, a total of 60 matches will be played. From these 60 matches, the three best-performing teams from each group (12 teams in total) will rise to the playoffs where seeding will take place. This will narrow down the number of teams to six, and using the same format, three teams will be determined. From these three teams, the top seed will have a guaranteed spot in the finals, but the other two teams will fight for the second spot with the second seed having a twice-to-beat advantage. Upon determining the two finalists, these teams will fight to ultimately determine the season's winner. With this, this new GSL season format increases the number of games required to determine the winner by about 53% - from 47 games to 71 or 72.

Essentially, the main difference between this format and the previous is the presence of a twice-to-beat determination factor that is brought about by the increase of teams leading to a "seeding" format with an odd number of teams at the end. Nonetheless, the grouping strategy still seems to be the most efficient way to execute everything together. For context, the number of games to be played if each team were to play all the other teams will be at 276. In contrast, the number of games to be played using this GSL Season Format is about 71 or 72 – a 70% decrease in the total number of games thus making the GSL as a whole more sustainable.

4.4 Accounting for External Factors

Essentially, with this new format, the primary factors that differentiate it with the initial model is the number of games along with the number of teams. However, in terms of external factors such as weather, climate, viewership, and time differences, the same algorithms utilized previously are still in place, just expanded to cater to the four additional teams.

In the context of both the weather and climate, the conditions of a given hemisphere within a given area dictate the timing of their matchups. For example, just as discussed in (3), those who reside in the Northern Hemisphere cannot play matches during the months between December and March due to the winter season being at its peak. Meanwhile, in the context of both viewership and time differences, matches where the participating teams are from areas with time differences of eight hours apart or more should be held on weekends. At the same time, the starting time of both matches should be within 8am to 8pm of both team's local times.

With the 50% increase in games however, the "buffer" times and gaps within the original schedule are to become much shorter with this team difference. The same 273-day season duration must now fit in 60 preliminary matches together with 11 or 12 playoff round matches. To attain the optimal results, the buffer time between the preliminary round and playoffs will be halved to a total of 7 days, and each of the following matches will then have a 7 day gap between them. Because of this, the duration of the playoffs (considering the worst case with 12 rounds) will take 7+12+(7*11)=96 days to complete. This is 9 days shorter than the initial schedule, but it is also the most optimal to ensure that all constraints are kept in place. This schedule also ensures that more time is placed on the number of preliminary matches which have now increased by 20.

With the playoffs taking 96 days to finish, there are 177 days to complete the 60 preliminary rounds. This means that each match will occur every $177 \div 60 \approx 2$ days (one day between each match). With this, the minimum and feasible recovery time of teams given this schedule is 7 days, so teams can only play every 4 matches.

This schedule, despite the shortened "buffer" times is still an organized and feasible one as possible gaps are still considered and all external factors are properly accounted for. In line with this, it keeps one important constraint continuous – no more than one planned match is played within one day.

5 | Generalizing the Model

5.1 Choosing the Teams

When choosing the teams, this part comes in two parts as discussed in (2): determining the number of teams per continent and choosing them based on a specific criteria. For the first part, the teams should be weighed depending on the total number of existing teams or associations for that given sport within that continent relative to the entire world. Once this is determined: the top associations per continent (based on this quota set by the first part) based on the presence of an airport and the number of active players set the number of teams for a season.

5.2 Determining the Format

5.2.1 The GSL Season Format

The GSL season format for any type of team sport game will remain the same. This is because this format is outcome based – it does not matter how the outcome (a win or a loss) was attained, what matters is the final result. Additionally, by proposing this format, it not only marks feasibility within the games, but it also provides sustainability by reducing the total number of games. At the same time, the matchup algorithms can still hold up. By considering both the location and timezone of a given area, the optimal groups and brackets can be formulated, and the best preliminary matches – leading to the better season outcomes – can be optimized.

5.2.2 Total Number of Games

Coupled with the format and schedule of the games, it is vital that the expected total number of games is calculated beforehand to better ensure adaptability and organization throughout the execution of the competition.

Following the seed format of the playoffs and semifinals, the total number of games to be played within a season can be computed as:

$$T_{games} = Preliminary\ Games\ +\ Playoffs\ +\ Semifinals\ +\ Finals\ +\ Game\ Determiners$$

This model simply adds all of the games required for all rounds of the games. However, it also includes the number of game determiners which are required to "bridge" the different parts of the round – especially when an odd number of teams result after a given round. As an example, if there were 18 teams sent to the playoffs, of those 18 teams, the top 9 will remain. This leads to problems in the number of matchups needed as not every team will have a respective pair. Thus, before the next round (with the Top 8), the bottom two teams will need to play a "game determiner" for the final spot in the semifinals. To balance this matchup however, the higher ranked team gets a twice to beat advantage. This means that for every game determiner sequence required, this can add either one or two games to the total number of games.

Using the above as a basis, a more expanded version of this expression is defined as:

$$T_{games} = \frac{kD_t}{2g_t}\left(\frac{D_t}{g_t} - 1\right) + \sum_{i=1}^{r}\left\lceil\frac{P_{i-1}}{2}\right\rceil + x_t\ where\ P_0 = \left\lceil\frac{D_t}{2}\right\rceil\ and\ r\ is\ the\ number\ of\ rounds\ to\ reduce\ P_0\ to\ 1.$$

The expressions used to derive this equation can be found in Appendix [3].

5.3 Scheduling the Teams

In the past few sections, we have already established that when creating a schedule for the GSL season matches, the external factors are vital constraints to hold with great care. In scheduling the teams, the external factors mainly time zones, viewership, and weather greatly impact the distribution and pacing of the schedule along with the corresponding "buffer" periods between each part of the season and each corresponding game.

Because of these factors, we obtain the following generalizations about the GSL season schedule when considering the various external factors:

1. Out of the total number of days, D_s, of a season, the total number of days allotted for playoffs is given as: $D_{po} = \sum_{i=1}^{r}\left[\frac{P_{i-1}}{2}\right] + x_t + b\left(x_t + \sum_{i=1}^{r}\left[\frac{P_{i-1}}{2}\right]\right)$ or $D_{po} = (b+1)\left(x_t + \sum_{i=1}^{r}\left[\frac{P_{i-1}}{2}\right]\right)$. This equation essentially adds all the playoff games – counted as days together with the product of the buffer period and the sum of game determiners and playoff games. This determines the number of days for games and the number of buffer period days given.
2. In line with this, for the preliminary round, the total number of days allotted, D_{pr}, is given as: $D_s - D_{po}$. From here, the buffer period and pacing for the preliminary round can be feasibly calculated. For the preliminary round, there will be a total of $\frac{kD_t}{2g_t}\left(\frac{D_t}{g_t} - 1\right)$ in total with D_{pr} days for implementation. Because of this, the suitable buffer period, to meet required constraints is given as: $b_p = \left[\frac{D_s - D_{po}}{\frac{kD_t}{2g_t}\left(\frac{D_t}{g_t} - 1\right)}\right] - 1$. Essentially, during the preliminary round, there are $\frac{kD_t}{2g_t}\left(\frac{D_t}{g_t} - 1\right)$ games to be played within D_{pr} days with at least b_p days between each matchup. An important thing to note, however, is that each team can only play another match when the number of games elapsed since their game, α, makes $\alpha(b_p - 1) \geq b$.

All of these considered, the external factors are properly met and set with these general constraints ,and a schedule meeting all required constraints is fulfilled.

6 | Conclusion

Several limitations were put in place to address the variation of events, travel routes, and match occurrences that exist at an international level. We limited country representation to existing national associations. Teams must be treated as a single figure due to the unpredictability of individuals and team preparation time.

Country location is represented by the location of their most active international airport, consequently requiring one from each participating country. This generalizes the transportation and travel of teams between countries to eliminate route variation for a replicable model. Seasonal variation was only considered between four regions, to solely account for the existence of four seasons and hemisphere behavior. Additionally, matches were treated as identical.. Match lengths were assumed to be negligible. Gender format and outdoor setting were assumed due to Ultimate culture, in which mixed gender is the most prominent format and most professional matches occur in the outdoor setting. The primary factor for the preliminary blocking was the location of the country's main airport without consideration of the established performance of an association in previous years as per the WFDF.

All of these limitations considered, the model is complete.

Team sports require rigid schedule consideration. The outcome of the schedule will greatly affect the lives of several participating athletes, and overall commitment is much less guaranteed. Our model is able to fully address the main factors that prioritize the convenience of both the athletes and the Global Sports League, and leave room for adaptable change that may stem from the prior commitments that these athletes may hold in relation to their already active national associations.

The strongest part of our model comes from the fact that we were able to consider the locations of each team as a tool to minimize travel time for all parties involved, and with this, our model keeps in mind the sustainability of both the league and the environment. This model is able to consider the time constraint of nine months while still putting the required schedule with minimized distances — not just for one team but for all — into action.

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Appendices

[1] Team groups generator (IMMCdemo2.py)

```
import pulp
import itertools
from math import radians, cos, sin, sqrt, atan2

teams = [
    {"name": "DFV (Germany)", "continent": "Europe", "lat": 50.0379, "lon": 8.5622},
    {"name": "UKU (UK)", "continent": "Europe", "lat": 51.470020, "lon": -0.454295},
    {"name": "FFFD (France)", "continent": "Europe", "lat": 49.009724, "lon": 2.547778},
    {"name": "FIG (Italy)", "continent": "Europe", "lat": 41.773540, "lon": 12.239712},
    {"name": "NFB (Netherlands)", "continent": "Europe", "lat": 52.308056, "lon": 4.764167},
    {"name": "SFF (Sweden)", "continent": "Europe", "lat": 59.65015, "lon": 17.94362},
    {"name": "OFSV (Austria)", "continent": "Europe", "lat": 48.11083, "lon": 16.57083},

    {"name": "JFDA (Japan)", "continent": "Asia", "lat": 35.553333, "lon": 139.781113},
    {"name": "CFDAC (China)", "continent": "Asia", "lat": 39.50917, "lon": 116.41056},
    {"name": "KUPA (South Korea)", "continent": "Asia", "lat": 37.463333, "lon": 126.440002},
    {"name": "UPAI (India)", "continent": "Asia", "lat": 28.556160, "lon": 77.100281},

    {"name": "SAFDA (South Africa)", "continent": "Africa", "lat": -26.134789, "lon": 28.240528},
    {"name": "KFDA (Kenya)", "continent": "Africa", "lat": -1.333731, "lon": 36.927109},
    {"name": "UFDF (Uganda)", "continent": "Africa", "lat": 0.044721, "lon": 32.443055},

    {"name": "USAU (USA)", "continent": "North America", "lat": 39.849312, "lon": -104.673828},
    {"name": "UC (Canada)", "continent": "North America", "lat": 43.677128, "lon": -79.633453},

    {"name": "FECODV (Colombia)", "continent": "South America", "lat": 4.7008, "lon": -74.1415},
    {"name": "FPD (Brazil)", "continent": "South America", "lat": -23.4258, "lon": -46.4683},

    {"name": "AFDA (Australia)", "continent": "Oceania", "lat": -33.947346, "lon": 151.179428},
    {"name": "NZU (New Zealand)", "continent": "Oceania", "lat": -37.0048, "lon": 174.7883},
]

def haversine(lat1, lon1, lat2, lon2):
    R = 6371
    dlat = radians(lat2 - lat1)
    dlon = radians(lon2 - lon1)
    a = sin(dlat / 2)**2 + cos(radians(lat1)) * cos(radians(lat2)) * sin(dlon / 2)**2
    c = 2 * atan2(sqrt(a), sqrt(1 - a))
    return R * c

n = len(teams)
distance = [[0]*n for _ in range(n)]
diversity = [[0]*n for _ in range(n)]

for i, j in itertools.combinations(range(n), 2):
    dist = haversine(teams[i]['lat'], teams[i]['lon'], teams[j]['lat'], teams[j]['lon'])
    distance[i][j] = distance[j][i] = dist
    diversity[i][j] = diversity[j][i] = 1 if teams[i]['continent'] != teams[j]['continent'] else 0

intercontinental_distances = []
for i in range(n):
    for j in range(i + 1, n):
        if diversity[i][j] == 1:
            intercontinental_distances.append(distance[i][j])

avg_intercontinental_distance = sum(intercontinental_distances) / len(intercontinental_distances)
print(f"Average intercontinental distance: {avg_intercontinental_distance:.2f} km")

#THIS IS THE PULP PROCEDURE:::
model = pulp.LpProblem("Team_Grouping", pulp.LpMinimize)
x = [[pulp.LpVariable(f"x_{i}_{j}", cat='Binary') for j in range(n)] for i in range(n)]

w_s = 1 #diStance
```

```
w_v = 10229.43 #diVersity
model += pulp.lpSum(w_s * distance[i][j] * x[i][j] - w_v * diversity[i][j] * x[i][j]
                    for i in range(n) for j in range(i+1, n))

for i in range(n):
    model += pulp.lpSum(x[i][j] for j in range(n) if i != j) == 4

for i in range(n):
    for j in range(i+1, n):
        model += x[i][j] == x[j][i]

for i, j, k in itertools.combinations(range(n), 3):
    model += x[i][j] + x[j][k] - 1 <= x[i][k]
    model += x[i][k] + x[k][j] - 1 <= x[i][j]
    model += x[j][i] + x[i][k] - 1 <= x[j][k]

model.solve(pulp.PULP_CBC_CMD(msg=0))
#ENDS HERE

groups = []
assigned = set()

for i in range(n):
    if i not in assigned:
        group = [i]
        for j in range(n):
            if i != j and pulp.value(x[i][j]) == 1 and j not in assigned:
                group.append(j)
        groups.append(group)
        assigned.update(group)

for idx, group in enumerate(groups, 1):
    print(f"Group {idx}:")
    for t in group:
        team = teams[t]
        print(f"{team['name']} ({team['continent']}, lat={team['lat']:.4f}, lon={team['lon']:.4f})")
    print()
```

Output:

```
Average intercontinental distance: 10229.43 km
Group 1:
DFV (Germany) (Europe, lat=50.0379, lon=8.5622)
SFF (Sweden) (Europe, lat=59.6501, lon=17.9436)
OFSV (Austria) (Europe, lat=48.1108, lon=16.5708)
UPAI (India) (Asia, lat=28.5562, lon=77.1003)
KFDA (Kenya) (Africa, lat=-1.3337, lon=36.9271)

Group 2:
UKU (UK) (Europe, lat=51.4700, lon=-0.4543)
NFB (Netherlands) (Europe, lat=52.3081, lon=4.7642)
USAU (USA) (North America, lat=39.8493, lon=-104.6738)
UC (Canada) (North America, lat=43.6771, lon=-79.6335)
FECODV (Colombia) (South America, lat=4.7008, lon=-74.1415)

Group 3:
FFFD (France) (Europe, lat=49.0097, lon=2.5478)
FIG (Italy) (Europe, lat=41.7735, lon=12.2397)
SAFDA (South Africa) (Africa, lat=-26.1348, lon=28.2405)
UFDF (Uganda) (Africa, lat=0.0447, lon=32.4431)
FPD (Brazil) (South America, lat=-23.4258, lon=-46.4683)

Group 4:
JFDA (Japan) (Asia, lat=35.5533, lon=139.7811)
CFDAC (China) (Asia, lat=39.5092, lon=116.4106)
KUPA (South Korea) (Asia, lat=37.4633, lon=126.4400)
AFDA (Australia) (Oceania, lat=-33.9473, lon=151.1794)
```

```
NZU (New Zealand) (Oceania, lat=-37.0048, lon=174.7883)
```

[2] Group matchups generator (IMMCmatchscheds.py)

```
import pulp
import itertools
from math import radians, cos, sin, sqrt, atan2

groups = {
    1: [
        {"name": "DFV (Germany)", "continent": "Europe", "lat": 50.0379, "lon": 8.5622},
        {"name": "SFF (Sweden)", "continent": "Europe", "lat": 59.6501, "lon": 17.9436},
        {"name": "OFSV (Austria)", "continent": "Europe", "lat": 48.1108, "lon": 16.5708},
        {"name": "UPAI (India)", "continent": "Asia", "lat": 28.5562, "lon": 77.1003},
        {"name": "KFDA (Kenya)", "continent": "Africa", "lat": -1.3337, "lon": 36.9271},
    ],
    2: [
        {"name": "UKU (UK)", "continent": "Europe", "lat": 51.4700, "lon": -0.4543},
        {"name": "NFB (Netherlands)", "continent": "Europe", "lat": 52.3081, "lon": 4.7642},
        {"name": "USAU (USA)", "continent": "North America", "lat": 39.8493, "lon": -104.6738},
        {"name": "UC (Canada)", "continent": "North America", "lat": 43.6771, "lon": -79.6335},
        {"name": "FECODV (Colombia)", "continent": "South America", "lat": 4.7008, "lon": -74.1415},
    ],
    3: [
        {"name": "FFFD (France)", "continent": "Europe", "lat": 49.0097, "lon": 2.5478},
        {"name": "FIG (Italy)", "continent": "Europe", "lat": 41.7735, "lon": 12.2397},
        {"name": "SAFDA (South Africa)", "continent": "Africa", "lat": -26.1348, "lon": 28.2405},
        {"name": "UFDF (Uganda)", "continent": "Africa", "lat": 0.0447, "lon": 32.4431},
        {"name": "FPD (Brazil)", "continent": "South America", "lat": -23.4258, "lon": -46.4683},
    ],
    4: [
        {"name": "JFDA (Japan)", "continent": "Asia", "lat": 35.5533, "lon": 139.7811},
        {"name": "CFDAC (China)", "continent": "Asia", "lat": 39.5092, "lon": 116.4106},
        {"name": "KUPA (South Korea)", "continent": "Asia", "lat": 37.4633, "lon": 126.4400},
        {"name": "AFDA (Australia)", "continent": "Oceania", "lat": -33.9473, "lon": 151.1794},
        {"name": "NZU (New Zealand)", "continent": "Oceania", "lat": -37.0048, "lon": 174.7883},
    ]
}

def haversine(lat1, lon1, lat2, lon2):
    R = 6371
    dlat = radians(lat2 - lat1)
    dlon = radians(lon2 - lon1)
    a = sin(dlat / 2)**2 + cos(radians(lat1)) * cos(radians(lat2)) * sin(dlon / 2)**2
    c = 2 * atan2(sqrt(a), sqrt(1 - a))
    return R * c

def schedule_group(group_teams):
    n = len(group_teams)
    distance = [[0]*n for _ in range(n)]

    for i, j in itertools.combinations(range(n), 2):
             dist = haversine(group_teams[i]['lat'], group_teams[i]['lon'], group_teams[j]['lat'],
group_teams[j]['lon'])
        distance[i][j] = distance[j][i] = dist

#THIS IS THE PULP PROCEDURE:::
    model = pulp.LpProblem("Group_Scheduling", pulp.LpMinimize)
     x = {(i, j): pulp.LpVariable(f"x_{i}_{j}", cat='Binary') for i in range(n) for j in range(n) if
i != j}

    model += pulp.lpSum(distance[i][j] * x[i, j] for i in range(n) for j in range(n) if i < j)

    for i in range(n):
        model += pulp.lpSum(x[i, j] for j in range(n) if i != j) == 2
```

```
        model += pulp.lpSum(x[j, i] for j in range(n) if i != j) == 2

    for i, j in itertools.combinations(range(n), 2):
        model += x[i, j] + x[j, i] == 1

    model.solve(pulp.PULP_CBC_CMD(msg=0))
#ENDS HERE

    print(f"Group Matches:")
    for i in range(n):
        for j in range(i + 1, n):
            if pulp.value(x.get((i, j), 0)) == 1:
                print(f"{group_teams[i]['name']} (Home) vs {group_teams[j]['name']} (Away)")
            elif pulp.value(x.get((j, i), 0)) == 1:
                print(f"{group_teams[j]['name']} (Home) vs {group_teams[i]['name']} (Away)")
    print()

for group_id, team_list in groups.items():
    schedule_group(team_list)ups.items():
    schedule_group(team_list))
```

Output:

```
Group Matches:
SFF (Sweden) (Home) vs DFV (Germany) (Away)
DFV (Germany) (Home) vs OFSV (Austria) (Away)
UPAI (India) (Home) vs DFV (Germany) (Away)
DFV (Germany) (Home) vs KFDA (Kenya) (Away)
OFSV (Austria) (Home) vs SFF (Sweden) (Away)
SFF (Sweden) (Home) vs UPAI (India) (Away)
KFDA (Kenya) (Home) vs SFF (Sweden) (Away)
UPAI (India) (Home) vs OFSV (Austria) (Away)
OFSV (Austria) (Home) vs KFDA (Kenya) (Away)
KFDA (Kenya) (Home) vs UPAI (India) (Away)

Group Matches:
NFB (Netherlands) (Home) vs UKU (UK) (Away)
USAU (USA) (Home) vs UKU (UK) (Away)
UKU (UK) (Home) vs UC (Canada) (Away)
UKU (UK) (Home) vs FECODV (Colombia) (Away)
USAU (USA) (Home) vs NFB (Netherlands) (Away)
UC (Canada) (Home) vs NFB (Netherlands) (Away)
NFB (Netherlands) (Home) vs FECODV (Colombia) (Away)
UC (Canada) (Home) vs USAU (USA) (Away)
FECODV (Colombia) (Home) vs USAU (USA) (Away)
FECODV (Colombia) (Home) vs UC (Canada) (Away)

Group Matches:
FIG (Italy) (Home) vs FFFD (France) (Away)
SAFDA (South Africa) (Home) vs FFFD (France) (Away)
FFFD (France) (Home) vs UFDF (Uganda) (Away)
FFFD (France) (Home) vs FPD (Brazil) (Away)
SAFDA (South Africa) (Home) vs FIG (Italy) (Away)
UFDF (Uganda) (Home) vs FIG (Italy) (Away)
FIG (Italy) (Home) vs FPD (Brazil) (Away)
UFDF (Uganda) (Home) vs SAFDA (South Africa) (Away)
FPD (Brazil) (Home) vs SAFDA (South Africa) (Away)
FPD (Brazil) (Home) vs UFDF (Uganda) (Away)

Group Matches:
CFDAC (China) (Home) vs JFDA (Japan) (Away)
KUPA (South Korea) (Home) vs JFDA (Japan) (Away)
JFDA (Japan) (Home) vs AFDA (Australia) (Away)
JFDA (Japan) (Home) vs NZU (New Zealand) (Away)
KUPA (South Korea) (Home) vs CFDAC (China) (Away)
AFDA (Australia) (Home) vs CFDAC (China) (Away)
CFDAC (China) (Home) vs NZU (New Zealand) (Away)
```

```
AFDA (Australia) (Home) vs KUPA (South Korea) (Away)
NZU (New Zealand) (Home) vs KUPA (South Korea) (Away)
NZU (New Zealand) (Home) vs AFDA (Australia) (Away)
```

[3] Derivation of T_{games}

Claim:

$$T_{games} = \frac{kD_t}{2g_t}\left(\frac{D_t}{g_t} - 1\right) + \sum_{i=1}^{r}\left[\frac{P_{i-1}}{2}\right] + x_t$$

To begin dissecting this equation, let us take a look at each part of the equation individually.

Given that $\frac{D_t}{g_t}$ represents the total number of participants within a single group for a variable, this first part of this equation shows how in each group, each participating team will be playing against all other participating teams within that group exactly once.

Hence, for each group, the total number of games played within that group is given by:

$$\frac{kD_t}{2g_t}\left(\frac{D_t}{g_t} - 1\right)$$

This format works because when grouping all the participating teams, one will have $\frac{D_t}{g_t}$ teams to "choose" from and $\left(\frac{D_t}{g_t} - 1\right)$ teams as a "choice" to go against the first "choice". This is then divided by two to remove all the repetitions in pairing all possible teams for matchups. Then, having k groups within the competition, this gets multiplied to get the total number of rounds during a preliminary round of a season.

Following the preliminary rounds, the playoffs, semifinals, and final rounds can now be calculated. Since the format of this round is in a round-robin style, in general, the total number of teams after each round gets halved until only one team, the winner, remains. Because of this, a summation of the floor of half the total number of teams per round should suffice in calculating the number of games until one team is left.

With this, the "additive value" x_t was added to the equation to serve as an increment to add upon for every time an odd valued $\left[\frac{P_{i-1}}{2}\right]$ arises.

This completes the derivation.

[4] Regions of Match Locations	
Matchup	Home Team Region
Group 1	
SFF (Sweden) vs DFV (Germany)	Northern Hemisphere
DFV (Germany) vs OFSV (Austria)	Northern Hemisphere

UPAI (India) vs DFV (Germany)	Tropical North
DFV (Germany) vs KFDA (Kenya)	Northern Hemisphere
OFSV (Austria) vs SFF (Sweden)	Northern Hemisphere
SFF (Sweden) vs UPAI (India)	Northern Hemisphere
KFDA (Kenya) vs SFF (Sweden)	Tropical North
UPAI (India) vs OFSV (Austria)	Tropical North
OFSV (Austria) vs KFDA (Kenya)	Northern Hemisphere
KFDA (Kenya) vs UPAI (India)	Tropical North
Group 2	
NFB (Netherlands) vs UKU (UK)	Northern Hemisphere
USAU (USA) vs UKU (UK)	Northern Hemisphere
UKU (UK) vs UC (Canada)	Northern Hemisphere
UKU (UK) vs FECODV (Colombia)	Northern Hemisphere
USAU (USA) vs NFB (Netherlands)	Northern Hemisphere
UC (Canada) vs NFB (Netherlands)	Northern Hemisphere
NFB (Netherlands) vs FECODV (Colombia)	Northern Hemisphere
UC (Canada) vs USAU (USA)	Northern Hemisphere
FECODV (Colombia) vs USAU (USA) Tropical North	Tropical North
FECODV (Colombia) vs UC (Canada) Tropical North	Tropical North
Group 3	
FIG (Italy) vs FFFD (France)	Northern Hemisphere
SAFDA (South Africa) vs FFFD (France)	Southern Hemisphere
FFFD (France) vs UFDF (Uganda)	Northern Hemisphere
FFFD (France) vs FPD (Brazil)	Northern Hemisphere
SAFDA (South Africa) vs FIG (Italy)	Southern Hemisphere
UFDF (Uganda) vs FIG (Italy)	Tropical North
FIG (Italy) vs FPD (Brazil)	Northern Hemisphere

UFDF (Uganda) vs SAFDA (South Africa)	Tropical North
FPD (Brazil) vs SAFDA (South Africa)	Tropical South
FPD (Brazil) vs UFDF (Uganda)	Tropical South
Group 4	
CFDAC (China) vs JFDA (Japan)	Northern Hemisphere
KUPA (South Korea) vs JFDA (Japan)	Northern Hemisphere
JFDA (Japan) vs AFDA (Australia)	Northern Hemisphere
JFDA (Japan) vs NZU (New Zealand)	Northern Hemisphere
KUPA (South Korea) vs CFDAC (China)	Northern Hemisphere
AFDA (Australia) vs CFDAC (China)	Southern Hemisphere
CFDAC (China) vs NZU (New Zealand)	Northern Hemisphere
AFDA (Australia) vs KUPA (South Korea)	Southern Hemisphere
NZU (New Zealand) vs KUPA (South Korea)	Southern Hemisphere
NZU (New Zealand) vs AFDA (Australia)	Southern Hemisphere

[5] Time Zones of Teams Home Country	
Team	Time Zone (UTC Offset)
DFV	UTC+2
UKU	UTC+1
FFFD	UTC+2
FIG	UTC+2
NFB	UTC+2
SFF	UTC+2
OFSV	UTC+2
JFDA	UTC+9
CFDAC	UTC+8
KUPA	UTC+9
UPAI	UTC+5:30

SAFDA	UTC+2
KFDA	UTC+3
UFDF	UTC+3
USAU	UTC−6
UC	UTC−4
FECODV	UTC−5
FPD	UTC−3
AFDA	UTC+10
NZU	UTC+12

[6] Matchup scheduler (IMMCtest.py)

```
from collections import defaultdict

matches = [
    {"match": "UFDF (Uganda) vs SAFDA (South Africa)", "region": "Tropical North"},
    {"match": "AFDA (Australia) vs KUPA (South Korea)", "region": "Southern Hemisphere"},
    {"match": "NZU (New Zealand) vs KUPA (South Korea)", "region": "Southern Hemisphere"},
    {"match": "NZU (New Zealand) vs AFDA (Australia)", "region": "Southern Hemisphere"},
    {"match": "UPAI (India) vs DFV (Germany)", "region": "Tropical North"},
    {"match": "AFDA (Australia) vs CFDAC (China)", "region": "Southern Hemisphere"},
    {"match": "SFF (Sweden) vs DFV (Germany)", "region": "Northern Hemisphere"},
    {"match": "DFV (Germany) vs OFSV (Austria)", "region": "Northern Hemisphere"},
    {"match": "DFV (Germany) vs KFDA (Kenya)", "region": "Northern Hemisphere"},
    {"match": "OFSV (Austria) vs SFF (Sweden)", "region": "Northern Hemisphere"},
    {"match": "SFF (Sweden) vs UPAI (India)", "region": "Northern Hemisphere"},
    {"match": "KFDA (Kenya) vs SFF (Sweden)", "region": "Tropical North"},
    {"match": "UPAI (India) vs OFSV (Austria)", "region": "Tropical North"},
    {"match": "OFSV (Austria) vs KFDA (Kenya)", "region": "Northern Hemisphere"},
    {"match": "KFDA (Kenya) vs UPAI (India)", "region": "Tropical North"},
    {"match": "NFB (Netherlands) vs UKU (UK)", "region": "Northern Hemisphere"},
    {"match": "USAU (USA) vs UKU (UK)", "region": "Northern Hemisphere"},
    {"match": "UKU (UK) vs UC (Canada)", "region": "Northern Hemisphere"},
    {"match": "UKU (UK) vs FECODV (Colombia)", "region": "Northern Hemisphere"},
    {"match": "USAU (USA) vs NFB (Netherlands)", "region": "Northern Hemisphere"},
    {"match": "UC (Canada) vs NFB (Netherlands)", "region": "Northern Hemisphere"},
    {"match": "NFB (Netherlands) vs FECODV (Colombia)", "region": "Northern Hemisphere"},
    {"match": "UC (Canada) vs USAU (USA)", "region": "Northern Hemisphere"},
    {"match": "FECODV (Colombia) vs USAU (USA)", "region": "Tropical North"},
    {"match": "FECODV (Colombia) vs UC (Canada)", "region": "Tropical North"},
    {"match": "FIG (Italy) vs FFFD (France)", "region": "Northern Hemisphere"},
    {"match": "SAFDA (South Africa) vs FFFD (France)", "region": "Southern Hemisphere"},
    {"match": "FFFD (France) vs UFDF (Uganda)", "region": "Northern Hemisphere"},
    {"match": "FFFD (France) vs FPD (Brazil)", "region": "Northern Hemisphere"},
    {"match": "SAFDA (South Africa) vs FIG (Italy)", "region": "Southern Hemisphere"},
    {"match": "UFDF (Uganda) vs FIG (Italy)", "region": "Tropical North"},
    {"match": "FIG (Italy) vs FPD (Brazil)", "region": "Northern Hemisphere"},
    {"match": "CFDAC (China) vs JFDA (Japan)", "region": "Northern Hemisphere"},
    {"match": "KUPA (South Korea) vs JFDA (Japan)", "region": "Northern Hemisphere"},
    {"match": "JFDA (Japan) vs AFDA (Australia)", "region": "Northern Hemisphere"},
    {"match": "JFDA (Japan) vs NZU (New Zealand)", "region": "Northern Hemisphere"},
    {"match": "KUPA (South Korea) vs CFDAC (China)", "region": "Northern Hemisphere"},
    {"match": "CFDAC (China) vs NZU (New Zealand)", "region": "Northern Hemisphere"},
```

```
    {"match": "FPD (Brazil) vs SAFDA (South Africa)", "region": "Tropical South"},
    {"match": "FPD (Brazil) vs UFDF (Uganda)", "region": "Tropical South"},
]

#idont want to split it manually
for match in matches:
    team1, team2 = match["match"].split(" vs ")
    match["team1"] = team1.strip()
    match["team2"] = team2.strip()
    if '(' in team1:
        match["home_location"] = team1.split('(')[1].strip(')')
    else:
        match["home_location"] = None
    match["scheduled"] = False

#constraints
team_availability = defaultdict(int)
schedule = {}

def is_available(team, index):
    return team not in team_availability or team_availability[team] + 12 < index

def is_valid_index(region, index):
    if region == 'Northern Hemisphere':
        return 32 <= index <= 168
    elif region == 'Southern Hemisphere':
        return 1 <= index <= 122
    elif region == 'Tropical North':
        return 1 <= index <= 61 or 154 <= index <= 168
    elif region == 'Tropical South':
        return 1 <= index <= 168
    return False

index = 1
while not all(m["scheduled"] for m in matches):
    found = False
    for m in matches:
        if not m["scheduled"]:
            t1, t2, r, loc = m["team1"], m["team2"], m["region"], m["home_location"]
             if is_valid_index(r, index) and is_available(t1, index) and is_available(t2, index) and
index not in schedule:
                schedule[index] = {"match": m["match"], "location": loc, "region": r}
                team_availability[t1] = index
                team_availability[t2] = index
                m["scheduled"] = True
                found = True
                break
    if found:
        index += 4
    else:
        index += 1
        if index > 200:
            print("your limits are too strict JDBHJWD")
            break

#scheduled
for idx in sorted(schedule.keys()):
    match = schedule[idx]
    print(f"index: {idx}")
    print(f"match: {match['match']}")
    print(f"location: {match['location']} ({match['region']})")
    print("-" * 20)

#unscheduled
unscheduled = [m["match"] for m in matches if not m["scheduled"]]
if unscheduled:
    print("\nmatches not scheduled:")
    for u in unscheduled:
```

```
        print(f"- {u}")
else:
    print("\nall matches were successfully scheduled YYIEEE")

print(f"\n# of scheduled matches: {len(schedule)}")
```

Output:

```
index: 1
match: UFDF (Uganda) vs SAFDA (South Africa)
location: Uganda (Tropical North)
--------------------
index: 5
match: AFDA (Australia) vs KUPA (South Korea)
location: Australia (Southern Hemisphere)
--------------------
index: 9
match: UPAI (India) vs DFV (Germany)
location: India (Tropical North)
--------------------
index: 13
match: KFDA (Kenya) vs SFF (Sweden)
location: Kenya (Tropical North)
--------------------
index: 17
match: FECODV (Colombia) vs USAU (USA)
location: Colombia (Tropical North)
--------------------
index: 21
match: NZU (New Zealand) vs KUPA (South Korea)
location: New Zealand (Southern Hemisphere)
--------------------
index: 25
match: AFDA (Australia) vs CFDAC (China)
location: Australia (Southern Hemisphere)
--------------------
index: 29
match: UPAI (India) vs OFSV (Austria)
location: India (Tropical North)
--------------------
index: 33
match: SFF (Sweden) vs DFV (Germany)
location: Sweden (Northern Hemisphere)
--------------------
index: 37
match: NFB (Netherlands) vs UKU (UK)
location: Netherlands (Northern Hemisphere)
--------------------
index: 41
match: NZU (New Zealand) vs AFDA (Australia)
location: New Zealand (Southern Hemisphere)
--------------------
index: 45
match: OFSV (Austria) vs KFDA (Kenya)
location: Austria (Northern Hemisphere)
--------------------
index: 49
match: SFF (Sweden) vs UPAI (India)
location: Sweden (Northern Hemisphere)
--------------------
index: 53
match: USAU (USA) vs UKU (UK)
location: USA (Northern Hemisphere)
--------------------
index: 57
match: UC (Canada) vs NFB (Netherlands)
location: Canada (Northern Hemisphere)
```

```
--------------------
index: 61
match: DFV (Germany) vs OFSV (Austria)
location: Germany (Northern Hemisphere)
--------------------
index: 65
match: FIG (Italy) vs FFFD (France)
location: Italy (Northern Hemisphere)
--------------------
index: 69
match: UKU (UK) vs FECODV (Colombia)
location: UK (Northern Hemisphere)
--------------------
index: 73
match: USAU (USA) vs NFB (Netherlands)
location: USA (Northern Hemisphere)
--------------------
index: 77
match: DFV (Germany) vs KFDA (Kenya)
location: Germany (Northern Hemisphere)
--------------------
index: 81
match: OFSV (Austria) vs SFF (Sweden)
location: Austria (Northern Hemisphere)
--------------------
index: 85
match: UKU (UK) vs UC (Canada)
location: UK (Northern Hemisphere)
--------------------
index: 89
match: NFB (Netherlands) vs FECODV (Colombia)
location: Netherlands (Northern Hemisphere)
--------------------
index: 93
match: SAFDA (South Africa) vs FFFD (France)
location: South Africa (Southern Hemisphere)
--------------------
index: 97
match: FIG (Italy) vs FPD (Brazil)
location: Italy (Northern Hemisphere)
--------------------
index: 101
match: UC (Canada) vs USAU (USA)
location: Canada (Northern Hemisphere)
--------------------
index: 105
match: CFDAC (China) vs JFDA (Japan)
location: China (Northern Hemisphere)
--------------------
index: 109
match: FFFD (France) vs UFDF (Uganda)
location: France (Northern Hemisphere)
--------------------
index: 113
match: SAFDA (South Africa) vs FIG (Italy)
location: South Africa (Southern Hemisphere)
--------------------
index: 118
match: KUPA (South Korea) vs JFDA (Japan)
location: South Korea (Northern Hemisphere)
--------------------
index: 122
match: FFFD (France) vs FPD (Brazil)
location: France (Northern Hemisphere)
--------------------
index: 126
match: CFDAC (China) vs NZU (New Zealand)
location: China (Northern Hemisphere)
```

```
--------------------
index: 131
match: JFDA (Japan) vs AFDA (Australia)
location: Japan (Northern Hemisphere)
--------------------
index: 135
match: FPD (Brazil) vs SAFDA (South Africa)
location: Brazil (Tropical South)
--------------------
index: 139
match: KUPA (South Korea) vs CFDAC (China)
location: South Korea (Northern Hemisphere)
--------------------
index: 144
match: JFDA (Japan) vs NZU (New Zealand)
location: Japan (Northern Hemisphere)
--------------------
index: 148
match: FPD (Brazil) vs UFDF (Uganda)
location: Brazil (Tropical South)
--------------------
index: 154
match: KFDA (Kenya) vs UPAI (India)
location: Kenya (Tropical North)
--------------------
index: 158
match: FECODV (Colombia) vs UC (Canada)
location: Colombia (Tropical North)
--------------------
index: 162
match: UFDF (Uganda) vs FIG (Italy)
location: Uganda (Tropical North)
--------------------

all matches were successfully scheduled YYIEEE

# of scheduled matches: 40
```

Report on Use of AI

1. OpenAI *Gemini* (Flash 2.0 version)

Query 1: can you outline the required pulp syntaxes for minimizing or maximizing a variable?

Output: Alright, let's outline the PuLP syntax for minimization and maximization, following a similar structure to how you would implement it in code:

Import the PuLP library:

Python

```
import pulp
```
This line makes the PuLP functions and classes available in your script.

Create an LpProblem instance:

Python

```
model = pulp.LpProblem("Problem_Name", pulp.LpMinimize) # For minimization
# OR
model = pulp.LpProblem("Problem_Name", pulp.LpMaximize) # For maximization
```
Here, you create an object representing your optimization problem. You provide a name for the problem (e.g., "Cost_Optimization", "Profit_Maximization"). The second argument specifies the objective: pulp.LpMinimize if you want to find the smallest possible value of your objective function, or pulp.LpMaximize if you want to find the largest.

Define your decision variables:

Python

```
variable_name = pulp.LpVariable("Variable_Description", lowBound=None, upBound=None, cat='Continuous')
# Examples:
x = pulp.LpVariable("Production_Quantity", lowBound=0, cat='Integer')
y = pulp.LpVariable("Is_Active", cat='Binary') # Binary variables are 0 or 1
z = pulp.LpVariable("Flow_Rate", lowBound=0) # Continuous with no upper bound
```
You create LpVariable objects for each decision you need to make. You provide a name that helps identify the variable in the output. You can also specify lowBound (minimum value), upBound (maximum value), and category ('Continuous' for real numbers, 'Integer' for whole numbers, 'Binary' for 0 or 1). If you don't specify bounds, they default to negative and positive infinity (for continuous and integer) or [0, 1] for binary.

Define the objective function:

Python

```
model += coefficient1 * variable1 + coefficient2 * variable2 + ... # For a linear objective
# Example for minimization:
model += 2 * x + 5 * y + 1.5 * z
# Example for maximization:
model += 10 * x - 3 * y + 7 * z
```
You define the function you want to minimize or maximize as a linear expression of your decision variables. You use the += operator to add this objective function to your LpProblem instance. The expression consists of the variables multiplied by their respective coefficients.

Add constraints (if any):

Python

```
model += coefficient_a * variable_a + coefficient_b * variable_b ... <= value_c
model += coefficient_d * variable_d + coefficient_e * variable_e ... >= value_f
model += coefficient_g * variable_g + coefficient_h * variable_h ... == value_i
# Examples:
model += x + y <= 100 # A capacity constraint
model += 3 * x - 2 * z >= 50 # A requirement constraint
model += y == 1 # Forcing a binary variable to a specific value
```
You add any restrictions or requirements as linear inequalities or equalities involving your decision variables. Again, you use the += operator to add these constraints to your LpProblem instance, using the operators <=, >=, and ==.

Solve the model:

Python

```
model.solve()
```

This line calls the default solver (or a solver you've specified) to find the optimal values for your decision variables that satisfy all the constraints and either minimize or maximize your objective function.

Access the results:

Python

```
print("Status:", pulp.LpStatus[model.status]) # Prints the status of the solution (e.g., "Optimal", "Infeasible")
print("Objective Value =", pulp.value(model.objective)) # Prints the optimal value of the objective function
for var in model.variables():
    print(var.name, "=", var.varValue) # Prints the optimal value of each decision variable
```

After solving, you can retrieve information about the solution, such as the status (whether an optimal solution was found), the optimal value of the objective function, and the optimal values of each decision variable.

In summary, the core syntax involves creating an LpProblem with the specified objective (minimize or maximize), defining LpVariable objects, formulating the objective function as a linear expression, adding constraints as linear expressions, solving the model using .solve(), and then accessing the results using pulp.value() and iterating through the variables.

Query 2: is there an error in my logic? the solver freezes when i run the program

```
model += pulp.lpSum(w_s * distance[i][j] * x[i][j] - w_v * diversity[i][j] * x[i][j]
             for i in range(n) for j in range(i+1, n))

for i in range(n):
    model += pulp.lpSum(x[i][j] for j in range(n)) == 4

for i in range(n):
    for j in range(i+1, n):
        model += x[i][j] == x[j][i]

for i, j, k in itertools.combinations(range(n), 3):
    model += x[i][j] + x[j][k] - 1 <= x[i][k]
    model += x[i][k] + x[k][j] - 1 <= x[i][j]
    model += x[j][i] + x[i][k] - 1 <= x[j][k]

model.solve(pulp.PULP_CBC_CMD(msg=0))
```

Output: Yes, there's a significant error in your logic that likely causes the solver to freeze or run for an extremely long time without finding a solution. The issue lies in how you're defining the constraint for the number of teams each team is grouped with.

Let's examine this constraint:

Python

```
for i in range(n):
    model += pulp.lpSum(x[i][j] for j in range(n)) == 4
```

Here, for each team i, you are summing x[i][j] for all j from 0 to n-1. Remember that x[i][i] will also be included in this sum. Since x[i][i] would logically represent a team being grouped with itself, and in your variable definition, you likely intend x[i][j] = 1 to mean team i is grouped with a different team j, this constraint is flawed.

The Correct Logic:

You want each team to be grouped with exactly four other teams. Therefore, you should exclude the case where i == j in your summation:

Python

```
for i in range(n):
    model += pulp.lpSum(x[i][j] for j in range(n) if i != j) == 4
```

Why your original constraint leads to problems:

If x[i][i] were allowed to be 1 (even though it doesn't make logical sense in the context of grouping with other teams), this constraint would try to force each team to have a "grouping" with itself included in the count of 4.

More importantly, the way you've set up your objective function and the subsequent group extraction logic implies that x[i][j] and x[j][i] represent the same pairing. By including x[i][i] in the sum and also having the symmetry constraint x[i][j] == x[j][i], you are overcounting the "connections" for each team.

For example, if team 0 is grouped with teams 1, 2, 3, and 4, then for i = 0, x[0][1], x[0][2], x[0][3], and x[0][4] would ideally be 1. Your original constraint would then include x[0][0] in the sum as well, potentially making it impossible to satisfy the equality with 4 unless x[0][0] is 0 and you have other non-zero x[0][j] values. This creates a very constrained and potentially infeasible problem, leading the solver to struggle.

In summary, change the constraint to exclude the case where i == j:

Python

```
for i in range(n):
    model += pulp.lpSum(x[i][j] for j in range(n) if i != j) == 4
```

This correction should significantly improve the performance of the solver and allow it to find a feasible solution (if one exists under your other constraints and objective).

Query 3: what's the use of binary decision variables?

Output: Binary decision variables are incredibly useful in mathematical optimization, including linear programming (which PuLP helps with), because they allow you to model "yes/no" or "on/off" decisions. They can only take on two values: 0 or 1.

One primary use is in selection problems. Imagine you have a set of items, and you need to decide which ones to include in a solution, perhaps due to a budget or capacity constraint. A binary variable can represent whether each item is selected (1) or not (0).

They are also crucial for modeling logical relationships between decisions. For instance, you might want to say "if we choose option A (variable = 1), then we must also choose option B (another variable = 1)." These "if-then," "either-or," and "and" conditions can be translated into mathematical inequalities involving binary variables.

Another key application is in representing fixed costs. If an activity is undertaken at all (represented by a binary variable being 1), a fixed cost is incurred. This allows for more realistic cost modeling.

In problems involving resource allocation, binary variables can indicate whether a specific resource is assigned to a particular task. Similarly, in scheduling, they can help determine the sequence or timing of activities.

Essentially, binary variables provide a way to incorporate discrete choices and logical constraints into mathematical optimization models, making it possible to solve problems that involve "yes/no" decisions and complex dependencies.

Query 4: What's the syntax for turning off the PuLP computation output of CBC solver?

Output: Here's the syntax to turn off the CBC solver's output:

Python

```
from pulp import LpProblem, LpMinimize, LpVariable, PULP_CBC_CMD

# Your PuLP problem definition
prob = LpProblem("My Optimization Problem", LpMinimize)
# ... your variables and constraints ...

# Solve using CBC and suppress output
prob.solve(solver=PULP_CBC_CMD(msg=False))

# Now, when you run this, you shouldn't see the detailed CBC solver output.
```

Explanation:

We import PULP_CBC_CMD from pulp. This allows us to explicitly specify the CBC solver and its options.
When calling prob.solve(), we pass the solver argument and set it to PULP_CBC_CMD(msg=False).
The msg=False part is the key! It tells the CBC solver to suppress its standard output during the solving process.
With this syntax, you should only see the final status of the solution (e.g., "Optimal") and the objective value, without the step-by-step output from the solver.

Let me know if you have any other questions! 😊

2. Google Sheet Smartfill AI *Auto-completed components of the schedule.*